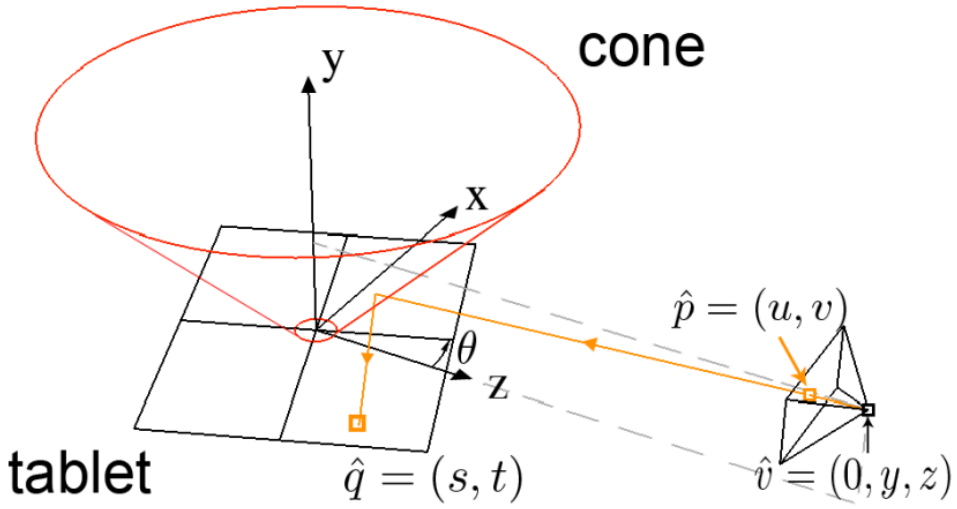
PDF of Paper
Link with video from researchers showing their results

My goal: jot down notes on this research group's steps to see if we can recreate them in a similar or (hopefully) less complex manner.

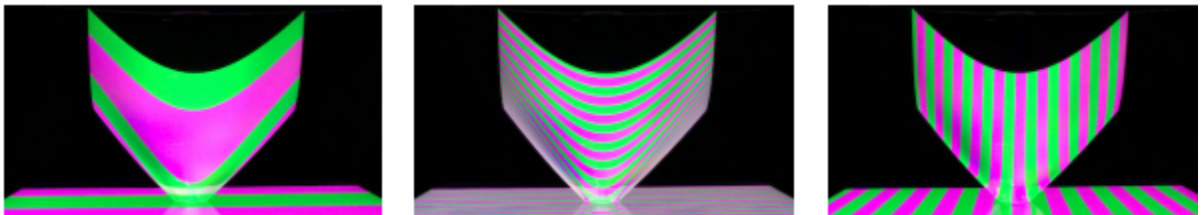
Distortion Calibration

- They wanted to produce a "perspective-correct image to someone who is viewing our display from the side".
    - This requires some calibration to determine how exactly to warp the image shown on the tablet to compensate for the distortion from a side angle



- They used this diagram to set up the variables/parameters for the equation used for their distortion.
- Their setup begins by stating
    - **e : p → q** is the function mapping points from the camera's plane p (on the right) to the tablet's plane of points q. They wanted to compute the inverse of that → **$f = e^{-1}$.**
    - This function **f** is referred to as the ***distortion map***.

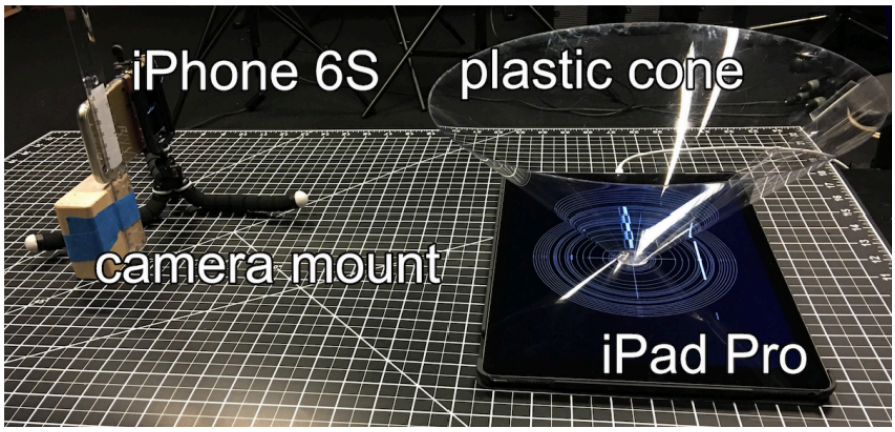
- To estimate the distortion map, they
    - Recorded a video of different highlighted patterns being projected from the tablet.



- Recovered discrete (not continuous) samples of e
  - Each green/magenta pixel represented either a 0 or a 1, allowing them to create a “binary mask” indicating which pixels from the projections on the tablet are being taken by the camera.
  - With this information, they were able to return a function for **e**, which they could then easily *inverse* to get a scattered version of $\mathbf{f} = \mathbf{e}^{-1}$.
- They then did some complex calculations to get a smoother function for f, shown here:

$$\mathbf{f}(\hat{q}) = \frac{\sum_{\hat{q}\in Q} w(\hat{p},\hat{q})\mathbf{f}(\hat{q})}{\sum_{\hat{q}\in Q} w(\hat{p},\hat{q})},\ w(\hat{p},\hat{q}) = \mathcal{N}(||\hat{p}-\hat{q}||_2^2 \mid \sigma).$$

- **My next steps**
  - With this info, I’m hoping I can work with Claude to figure out how we can replicate the setup methodology so we can feed it data to create its own **f(q^)** function. It can use this function to more properly and precisely warp images on the screen.
  - The hardest part of this would most likely be figuring out what point of view to use for the calculations. To be more specific



The researchers had their iPhone at a fixed distance from the cone and at a pretty straight on and level view. **Students in a classroom will be a lower vantage point & pretty far away.**